Week 7 Lectures 1 & 2

STAT2004 / 2904 / 7004

Announcements:

- Assignment 2 due Fri 11 Sep @ 4pm
- Assignment 3 release on Mon 14 Sep @ 4pm
- Alan's consultation hour Fridays 12–1pm or via appt

Review of STAT1201/1301/2004a

Consider a large popⁿ of units. We are interested in some numerical property of this population (i.e, **parameter**). This could be an average measurement μ in the popⁿ (eg., average weight, blood pressure, income, etc...) or a popⁿ proportion p of a certain characteristic (e.g., % of left-handed students at UQ, % of Australian adults with a certain disease, etc...)

Suppose a random sample of n units is taken from our population, & a sample **statistic** is computed. This could be a sample mean $\bar{X}$ for quantitative variables, or a sample proportion $\hat{p}$ for a categorical variable, etc...

The burning question is: How confident can we be that our sample statistic is close to our popⁿ parameter?

DEMO

Skittles = 23
M&Ms = 24

In our demo, the sample size was $n = 47$, & our point estimate was $\hat{p} = 23/47 = 0.489 = 48.9\%$.

How confident can we be that this point estimate is close to the popⁿ proportion?

§ 6 Margins of error & level of confidence

"closeness" can be formalized as a margin of error
"confidence" can be formalized as a level of confidence.

Recall from our first year stats, that an (approx) 90%, 95% or 99% confidence interval for p can be constructed as:

$$90\%\ CI = \hat{p} \pm 1.64 \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$\hat{p}$: point estimate; 1.64: critical value; $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$: s.e

$1.64 \times$ s.e: 90% margin of error

$$95\%\ CI : \hat{p} \pm 1.96 \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$99\%\ CI : \hat{p} \pm 2.58 \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

99%: level of confidence; $2.58 \times$ s.e: margin of error.

From our particular sample, we had $n = 47$, $\hat{p} = 0.489$ & so we enumerate our 90% CI as:

$$90\%\ CI = 0.489 \pm 1.64 \sqrt{\frac{0.489 \times 0.511}{47}}$$

$$= 0.489 \pm 1.64 \times 0.073$$

$$= 0.489 \pm 0.120$$

$$= 0.369 \text{ to } 0.609.$$

$$= 36.9\% \text{ to } 60.9\%$$

Two questions:

1. We know that the critical values (eg. 1.64, 1.96 & 2.58) come from a standard normal dist^n, BUT what does the standard normal have to do with proportions?
2. Why do these critical values NOT depend on the underlying true parameter value p?

Answer to both of these questions lies with the concept of a pivot quantity / pivot variable / pivot.

Defn: A pivot variable is a theoretical (& generally non-computable) transformation of a random variable (data) such that its dist^n no longer depends on any unknown parameters

Eg. 6.2 (Normal data). Let $X_1, X_2, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ where μ is the unknown mean, but assume σ^2 is known for now (for simplicity).

What do we know about $\bar{X} = \frac{X_1 + X_2 + \ldots + X_n}{n}$?

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

Thus, the theoretical (& non-computable) transformation

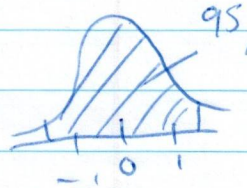
$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

no longer depends on any unknown parameters.

Hence, $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ is a pivot. □

Pivot variables can be used to construct confidence intervals & hypothesis tests for pop^n parameters.

Eg. 6.2 (normal data, continued)



We can start with a prob statement:

$$95\% = P\left(-1.96 \leq N(0,1) \leq +1.96\right)$$

purely a known prob. statement about standard normal.

pivot

$$= P\left(-1.96 \leq \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \leq +1.96\right)$$

$$= P\left(-1.96\,\sigma/\sqrt{n} \leq \bar{X}-\mu \leq +1.96\,\sigma/\sqrt{n}\right)$$

$$= P\left(\bar{X}+1.96\,\sigma/\sqrt{n} \geq \mu \geq \bar{X}-1.96\,\sigma/\sqrt{n}\right)$$

This is how the classical 95% CI for μ is derived:

$$\bar{X} \pm 1.96\,\sigma/\sqrt{n}$$

NOTE: What is random in our probability statement (& what is fixed)? Because $\bar{X}$ is random (changes from sample to sample), the end points are random. BUT the pop^n mean μ is fixed.

We can be 95% confident that this $\bar{X} \pm 1.96\,\sigma/\sqrt{n}$ contains the true pop^n mean μ, because this random procedure works 95% of the time (95% of samples).

For this reason, it is helpful to think of confidence interval procedures, because we are confident in the procedure & not in the particular numbers that we happen to get.

So how confident can we be that the sample proportion of $\hat{p} = 0.489$ is exactly the pop^n parameter p?

0% confident.

This concept of pivots is even more useful for more complex/realistic scenarios, such as when both pop^n mean μ & pop^n variance σ^2 are unknown.

Confidence interval for μ when σ^2 is also unknown

Let $X_1, X_2, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ where both μ & σ^2 are unknown.

We already know that the MLE for μ is

(1) $$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n} \sim N(\mu, \sigma^2/n)$$

We have also seen that (bias-corrected) MLE of σ^2

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})^2$$

which has dist^n

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

(2) $$\Leftrightarrow \quad S^2/\sigma^2 \sim \frac{\chi^2_{(n-1)}}{n-1}$$

③ Moreover (!), $\bar{X}$ & S^2 are independent (!!)

Let's consider the transformation given by:

$$T = \frac{\bar{X}-\mu}{S/\sqrt{n}} = \frac{\left[\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\right]}{\left[S/\sigma\right]} \sim \frac{N(0,1)}{\sqrt{\frac{\chi^2_{(n-1)}}{n-1}}} \quad \begin{matrix}① \\ ③ \\ ②\end{matrix}$$

indep

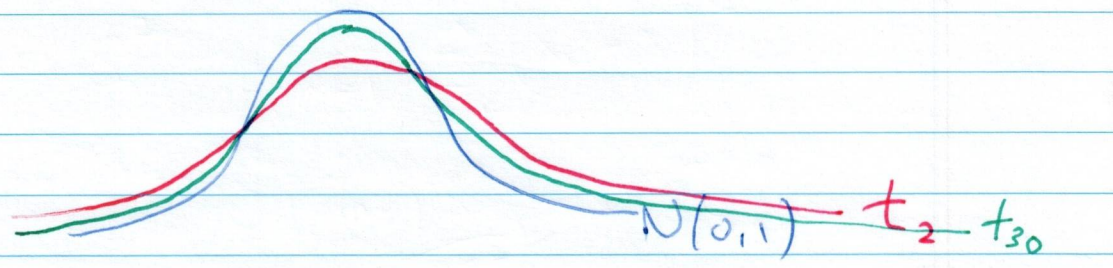
$$\equiv T_{n-1}$$

by def of a t-distn.

(see student 1908 paper for a proof by brute force)

Thus, T is a theoretical (but non-computable) transformation of our data that has a distn that no longer depends on any unknown parameters. i.e. T is a pivot.

Like a standard normal, t's are also symmetric & somewhat bell-curved:



To construct a $(1-\alpha)100\%$ CI for μ when σ^2 is also unknown, we again start with a fully known probability statement:

(fully known prob. statement)

$$(1-\alpha)100\% = P\left(-t^{(1-\alpha/2)} \leq T_{n-1} \leq t^{(1-\alpha/2)}\right)$$

T_{n-1} pivot

$$= P\left(-t^{(1-\alpha/2)} \leq \frac{\bar{X}-\mu}{S/\sqrt{n}} \leq t^{(1-\alpha/2)}\right)$$

$$\vdots$$

$$= P\left(\bar{X} + t^{(1-\alpha/2)} S/\sqrt{n} \geq \mu \geq \bar{X} - t^{(1-\alpha/2)} S/\sqrt{n}\right)$$

This is how the familiar $\left(\bar{X} \pm t^{(1-\alpha/2)} S/\sqrt{n}\right)$ confidence interval is formally derived.

How do we find the cutoffs $t^{(1-\alpha/2)}$? We can use tables, but not every table will have every combination of degrees-of-freedom & confidence level. So much better to use software.

For example, for sample size $n = 42$ $(df = 41)$ & confidence level of 97%, say. Then what is $t_{41}^{(0.985)}$

```
> qt(0.985, 41)
[1] 2.248
```

So a 97% CI for μ when σ^2 is unknown using $n = 42$, would be:

$$\bar{X} \pm 2.248\, S/\sqrt{42}$$

Not all dist's are symmetric, so not all CIs are symmetric. In fact, not all CIs are additive ($\pm$) in their margin of error. Some CIs are multiplicative (i.e., relative) in their margins of error.

Confidence intervals for a pop^n variance σ^2

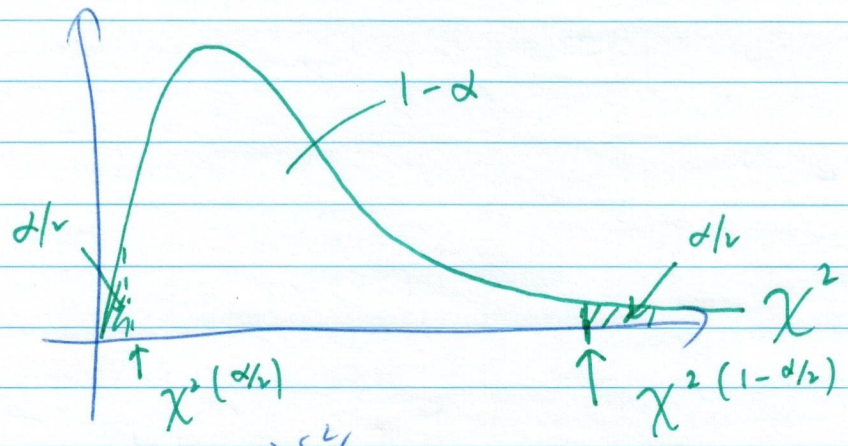
Let $X_1, X_2, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ where μ & σ^2 are both unknown. We have already seen that

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{(n-1)} \qquad \text{where } S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

↑ does NOT depend on any unknown parameters.

& so $(n-1)S^2/\sigma^2$

χ^2 dist's are NOT symmetric, & look like this:



Because $(n-1)S^2/\sigma^2$ is a pivot, we can again start with a fully known prob. statement: fully known prob statement

$$(1-\alpha)100\% = P\left(\chi^2_{(\alpha/2)} \le \chi^2_{(n-1)} \le \chi^2_{(1-\alpha/2)}\right)$$

pivot

$$P\left(\chi^2_{(\alpha/2)} \le (n-1)S^2/\sigma^2 \le \chi^2_{(1-\alpha/2)}\right)$$

$$= P\left(\frac{(n-1)S^2}{\chi^2_{(\alpha/2)}} \geq \sigma^2 \geq \frac{(n-1)S^2}{\chi^2_{(1-\alpha/2)}} \right)$$

So a $(1-\alpha)100\%$ CI for σ^2 can be constructed as

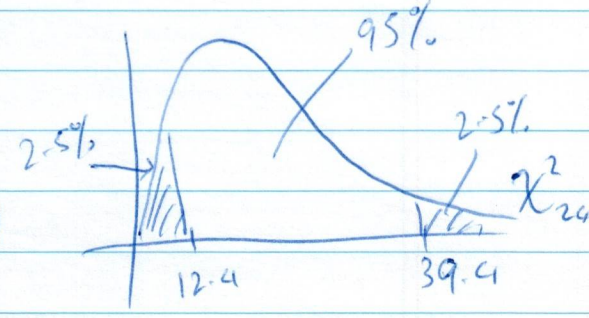
$$\left(\frac{(n-1)S^2}{\chi^2_{(n-1)(1-\alpha/2)}} , \frac{(n-1)S^2}{\chi^2_{(n-1)(\alpha/2)}} \right)$$

Example 6.4 On the label of a certain pack of aspirin, it is written that the sd of the tablet's weight is 1.0mg.

To investigate this, we take a sample of $n = 25$ tablets & computed a sample sd of $S = 1.3$mg. Construct a 95% CI for σ^2 (& hence for σ)

The sample size here is $n = 25$, & so $df = 24$ for estimating σ^2. We want to find the lower 2.5% tile & upper 2.5% tile of a χ^2_{24} distn. Using R:

```
> qchisq(0.025, 24)
[1] 12.4
> qchisq(0.975, 24)
[1] 39.4
```



So a 95% CI for σ^2 can be constructed as

$$\left(\frac{(n-1)S^2}{39.4} , \frac{(n-1)S^2}{12.4} \right) = \left(\frac{24 \times 1.3^2}{39.4} , \frac{24 \times 1.3^2}{12.4} \right)$$

$$= (1.04, 3.27)\ mg^2$$

for pop variance σ^2

We can convert this to 95% CI for σ by taking $\sqrt{}$:

$$\left(\sqrt{1.04}, \sqrt{3.27}\right) = (1.02, 1.81) \text{ mg}$$

contains $S = 1.3$, but it is NOT symmetric around it.

Interpret: We can be 95% confident that the true σ of the manufacturing process is between 1.02 mg & 1.81 mg.

Philosophy corner: Can we ever say something like "there's a 95% chance that the true value is between xxx & yyy?"

From a frequentist interpretation of probability, NO.
From a Bayesian interpretation of probability, Maybe/Yes

Final question: So, what do you think of the manufacturer's claim that $\sigma = 1.0$ mg

Coming back to the original experiment with skittles & M&Ms; can we construct pivots for discrete random variables?

Typically, NO. Because transformations of a discrete r.v. typically change the support of the dist:

Suppose $X \sim Bin(n=5, p=0.4)$

then X can take on values 0, 1, 2, 3, 4, 5.

We know that $E(X) = np = 2$

& $Var(X) = np(1-p) = 5 \times 0.4 \times 0.6 = 1.2$

& so $\sqrt{Var(X)} = \sqrt{1.2} = 1.095$

$X - E(X)$ takes on values $-2, -1, 0, 1, 2, 3$

$$\frac{X - E(X)}{sd(X)} = \frac{-2}{1.095}, \frac{-1}{1.095}, 0, \frac{1}{1.095}, \frac{2}{1.095}, \frac{3}{1.09}$$

??

So the dist^n will always still depend on unknown parameters.

So in general, there is NO exact pivot for discrete random variables. However, we can aim for approximate pivots (asymptotic pivots) by appealing to limit theorems for when we have large enough sample sizes.